

GO-FOC SERIES ESC -GENERAL OPERATION MANUAL

Rev. V2025.8.12

WARNING

Read this manual thoroughly before operating the device. Always follow the specified operating procedures.

Unauthorized modifications or operation beyond the rated limits may result in serious injury, death, or property damage.

ELECTRICAL WARNING

This product supports high-voltage and high-current input, which can be hazardous.

Before powering on, ensure all wiring is correct and secure, and that proper insulation measures are in place. Always install a fuse on the input side to prevent extreme failure conditions. Use of this module requires basic knowledge of electronics. If you are not experienced, proceed with caution.

This product is developed on the VESC platform.

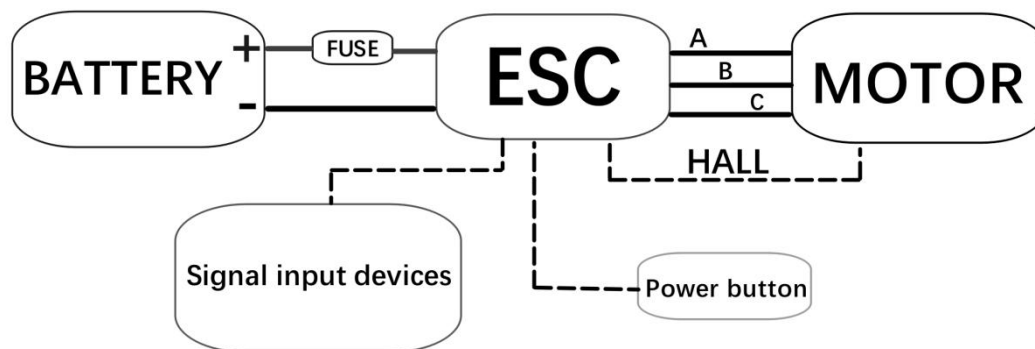
VESC Official Website : <https://vesc-project.com/>

Please consult the product page on the website for ESC parameters :

<https://www.makerx-tech.com/collections/x-esc>

Please read before use: <https://www.makerx-tech.com/pages/faq>

HARDWARE WIRING:



CONNECTING THE BATTERY AND MOTOR:

A / B / C: Motor phase wires

+ / -: Battery positive (+) and negative (-) terminals

Always use the motor and battery compatible with the specific ESC model. Select wire gauge appropriately according to the application and the anticipated current load.

INPUT HARDWARE WIRING

ADC Input :

Throttle Signal Output	→	ADC1 (ESC)
Brake Lever or Button	→	ADC2 (ESC)
Throttle Grip 5V	→	5V (ESC)
Throttle Grip GND	→	GND (ESC)

UART Receiver Input:

Receiver TX	→	RX (ESC)
Receiver RX	→	TX (ESC)
Receiver 5V	→	5V (ESC)
Receiver GND	→	GND (ESC)

Motor Hall Sensor Input:

HALL 5V	→	5V (ESC)
Motor Temperature	→	Temp (ESC)
HALL HA	→	HA (ESC)
HALL HB	→	HB (ESC)
HALL HC	→	HC (ESC)
HALL GND	→	GND (ESC)

UART Bluetooth Module:

Bluetooth Module TX	→	RX/RX2 (ESC)
Bluetooth Module RX	→	TX/TX2 (ESC)
Bluetooth Module 5V	→	5V (ESC)
Bluetooth Module GND	→	GND (ESC)

PWM Receiver Input

Receiver PPM	→	PPM (ESC)
Receiver 5V	→	5V (ESC)
Receiver GND	→	GND (ESC)

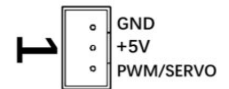
CAN Connection

CANH (VESC1)	→	CANH (VESC2)
CANL (VESC1)	→	CANL (VESC2)

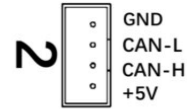
Ensure all VESCs share a common battery negative.

PORT DESCRIPTION

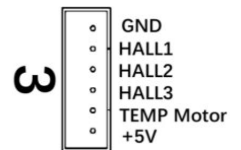
1 PWM: Input devices, such as 2.4GHz transceivers, connect via Pulse Position Modulation (PPM) to control the motor's output power and speed.



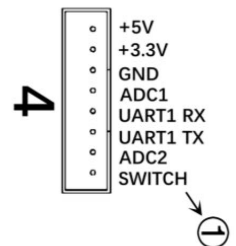
2 CAN: The CAN bus facilitates communication among VESC devices arranged in an array. Additionally, it serves as a universal bus to connect VESC controllers with other CAN-enabled devices.



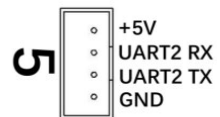
3 SENSOR: Sensor ports for ABI, HALL, AS5047P motor encoders, and other encoders listed in the VESC Tool. The TEMP port is used for motor temperature detection.



4 COMM: This port features I2C, SPI, UART, and ADC interfaces, enabling communication with external devices such as microcontrollers (e.g., Arduino, Raspberry Pi) or analog input devices including analog throttles and switches.



5 UART2: The secondary UART port is available for connecting devices like Bluetooth modules, VESC displays, and other peripherals.



6 USB: The USB port is used to connect to a computer for configuration, firmware updates, and real-time data analysis.



Note: Some silkscreen markings are simplified, for example, ADC1 is marked as A1. Please refer to the actual markings on the ESC.

① SWITCH Switch Button Accessory: Most Go-FOC series ESCs support the switch button function, allowing the device to be set into ultra-low power sleep mode.

INDICATOR LIGHTS:

- **Blue:** VESC controller is powered on.
- **Green (dim):** The device has the correct firmware installed and is running.
- **Green (bright):** The controller is actively driving the motor.
- **Red:** Fault code detected. Refer to VESC TOOL for details on the fault.

1. SOFTWARE PREPARATION

1. Download VESC Tool:

https://vesc-project.com/vesc_tool

2

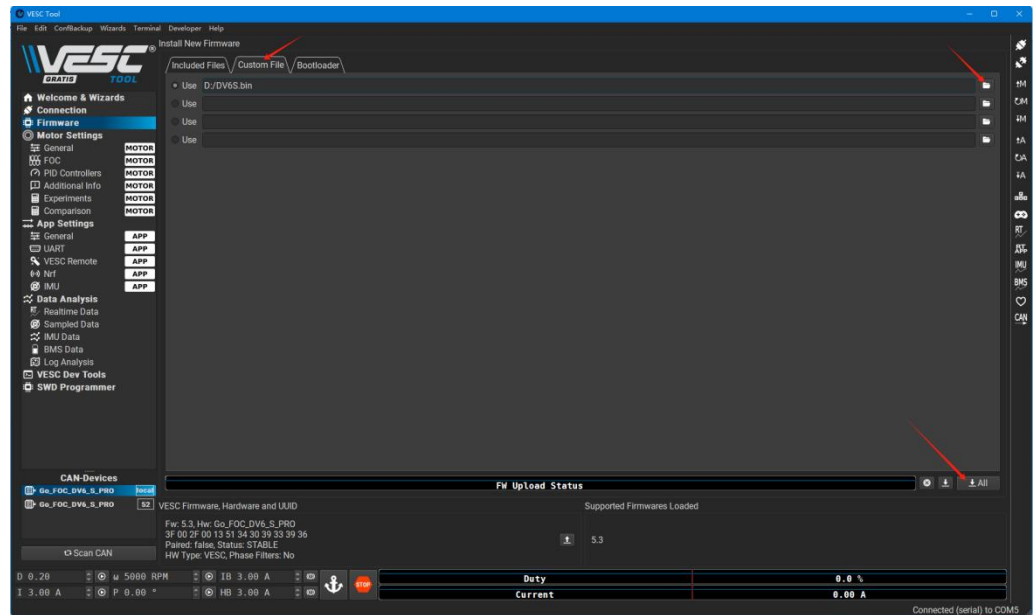
After ensuring proper hardware connections, power on the device and connect to the VESC Tool on your computer (via USB) or the mobile app (via Bluetooth) to configure the ESC.

Note: The VESC hardware firmware version must correspond to the correct VESC TOOL version:

For example, ESC firmware 5.02 corresponds to VESC TOOL version 3.00;

ESC firmware 5.03 corresponds to VESC TOOL version 3.01;

ESC firmware 6.05 corresponds to VESC TOOL version 6.05.



2. FIRMWARE UPGRADE:

1

First, upgrade the Bootloader.

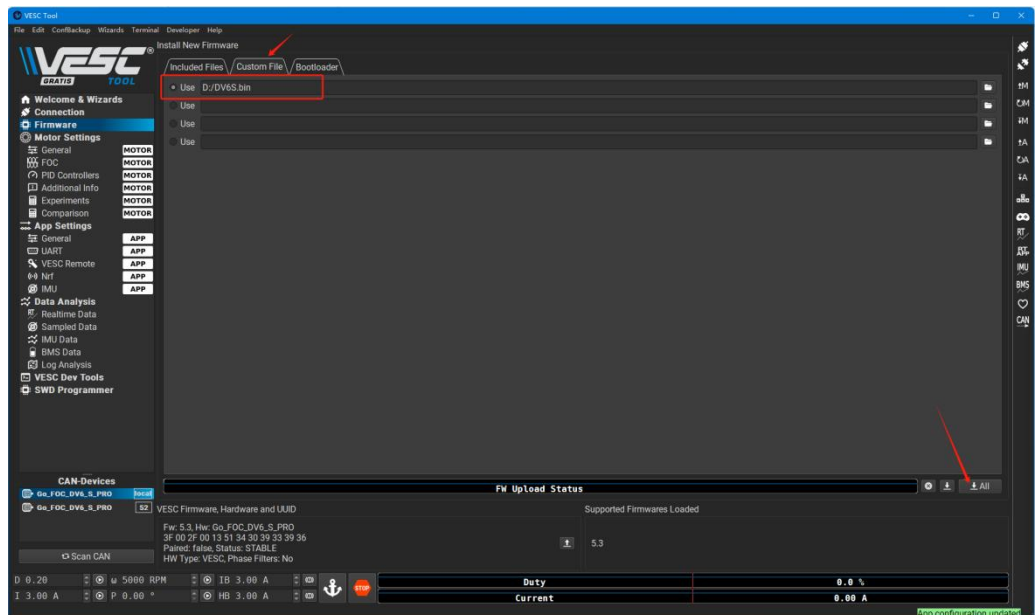
2

Select the corresponding firmware file and click “ALL.”

Firmware file download:

<https://www.makerx-tech.com/pages/download>

In most cases, you do not need to manually upgrade the firmware, as a stable version is already installed at the factory.



3. FOC

CONFIGURATION:

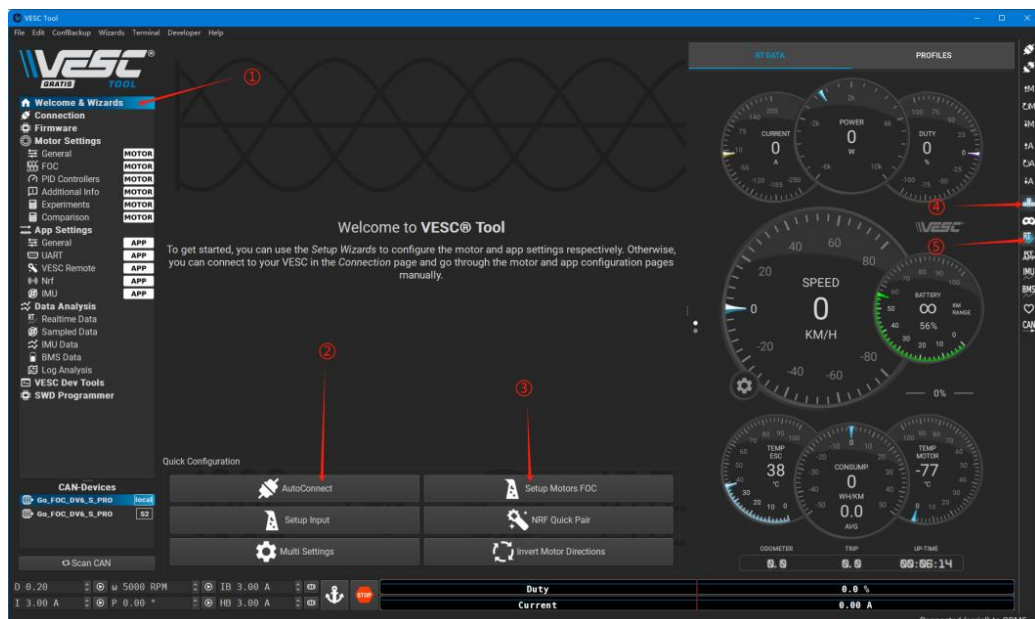
1. Enter the wizard page.

2. Click here to connect to the ESC.

3. Click this button to enter motor FOC configuration.

4. Turn on this switch to test the motor function using the keyboard.

5. Turn on this button to view the ESC's real-time data.

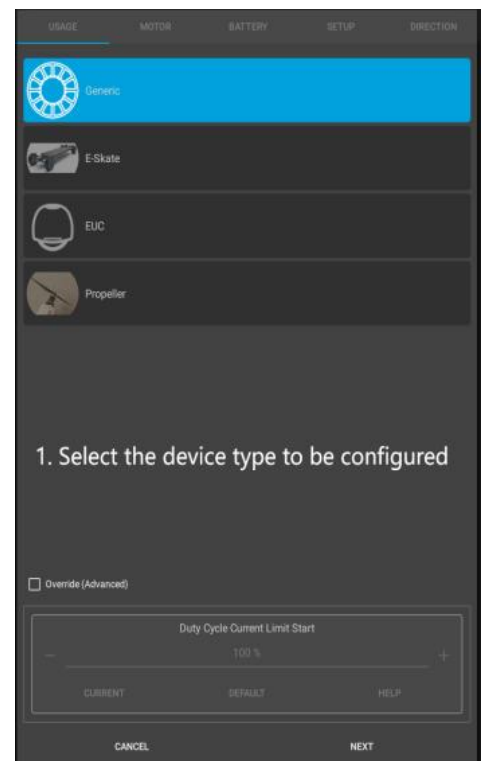
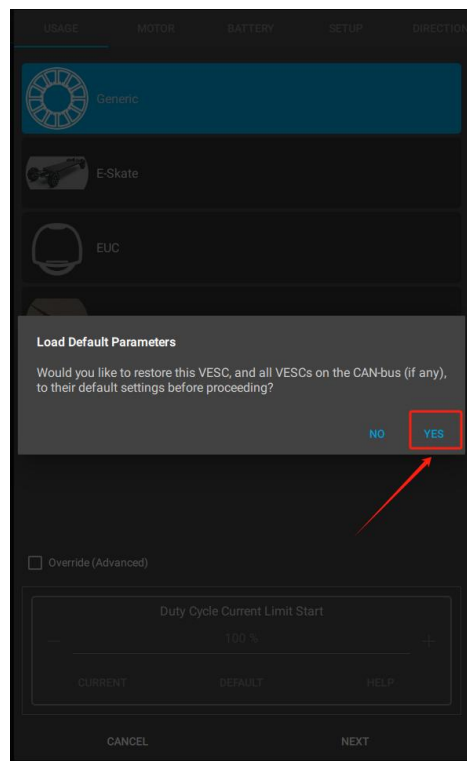


It is recommended to restore the default parameters each time you perform FOC configuration.

In the dialog box shown, select **YES**.

1

Select the device type to be configured. Then click Next.



2

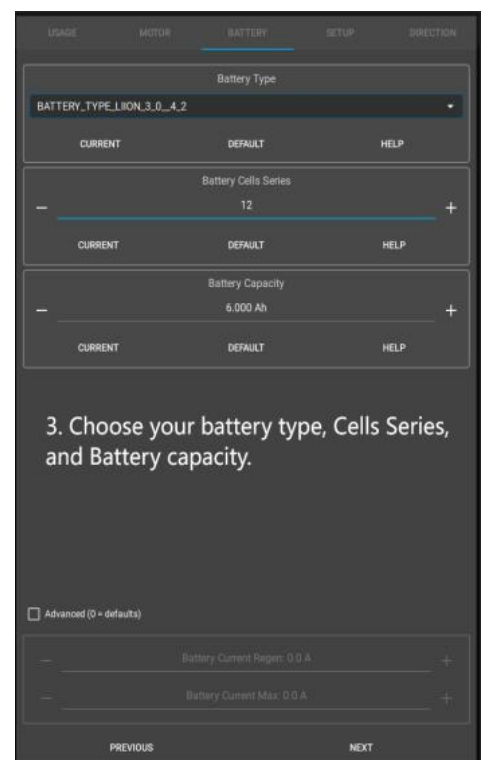
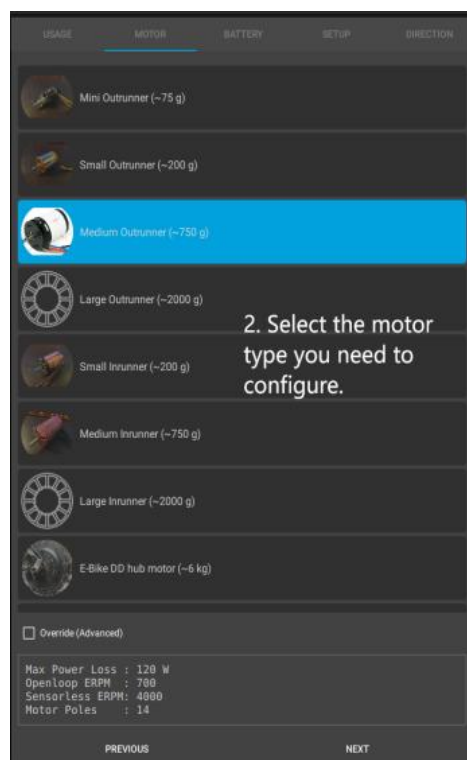
Select the type of motor you want to configure, such as outrunner, inrunner, weight, etc.

Then click Next.

3

Select your battery type, series count, and capacity.

Then click Next.



4

Configure your gearbox, drive pulley, driven pulley, and wheel diameter. After setup, you will get highly accurate feedback in the VESC Tool application, including speed, estimated range, energy consumption per kilometer, and more

5

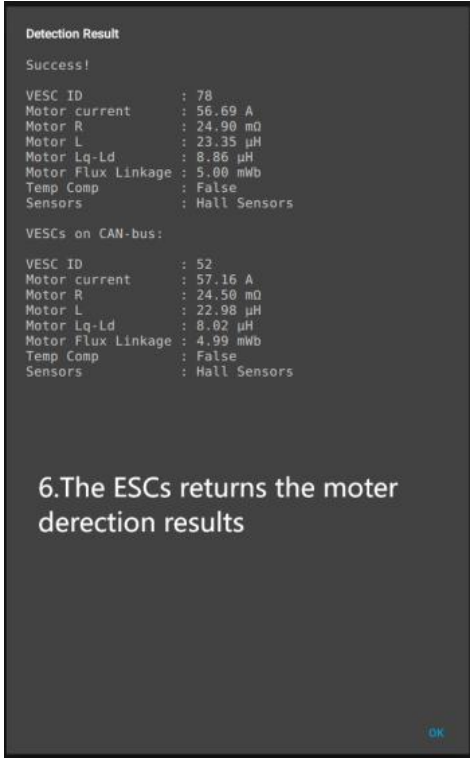
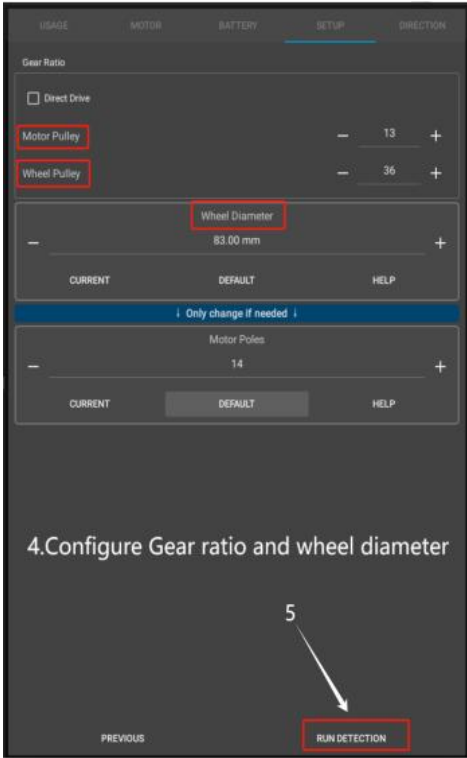
Click **RUN DETECTION** to start motor self-detection.

Note: The motor will rotate. Please ensure it is not obstructed, disconnect any load, and observe safety precautions!

6

Once motor detection is complete and results are returned, click OK to automatically write the results to the ESC.

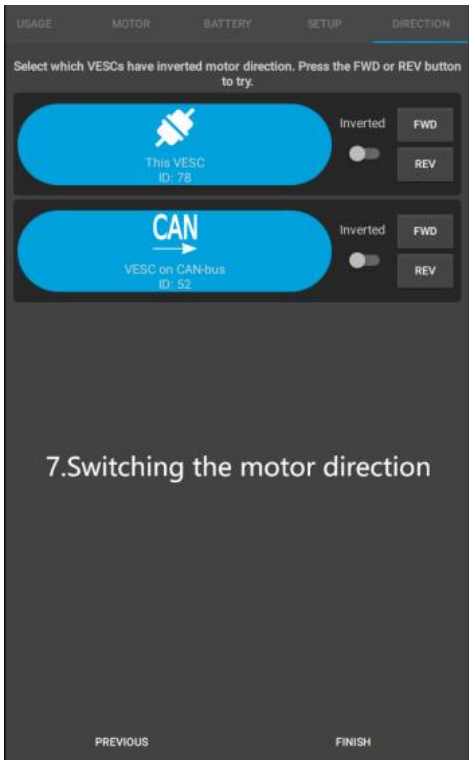
Note: Please check the returned data for abnormalities, such as certain parameters being unusually large or zero.
Running the ESC with incorrect data may cause damage.



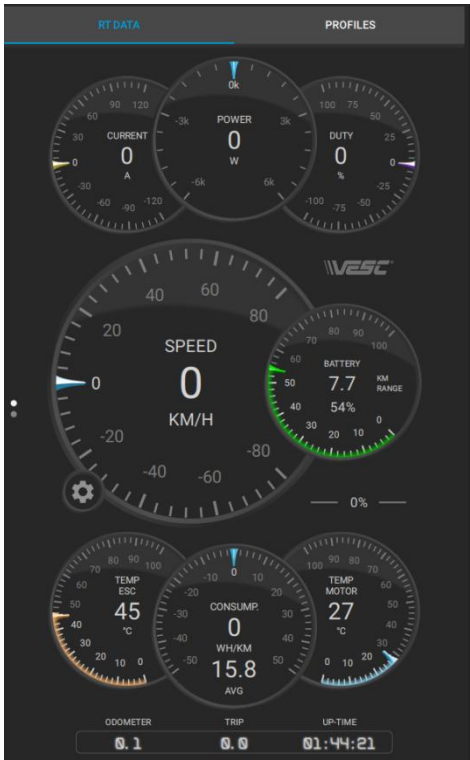
6.The ESCs returns the moter derection results

7

To change the motor direction, click Inverted to switch the motor's rotation direction. Then click the Finish button to complete the FOC configuration.

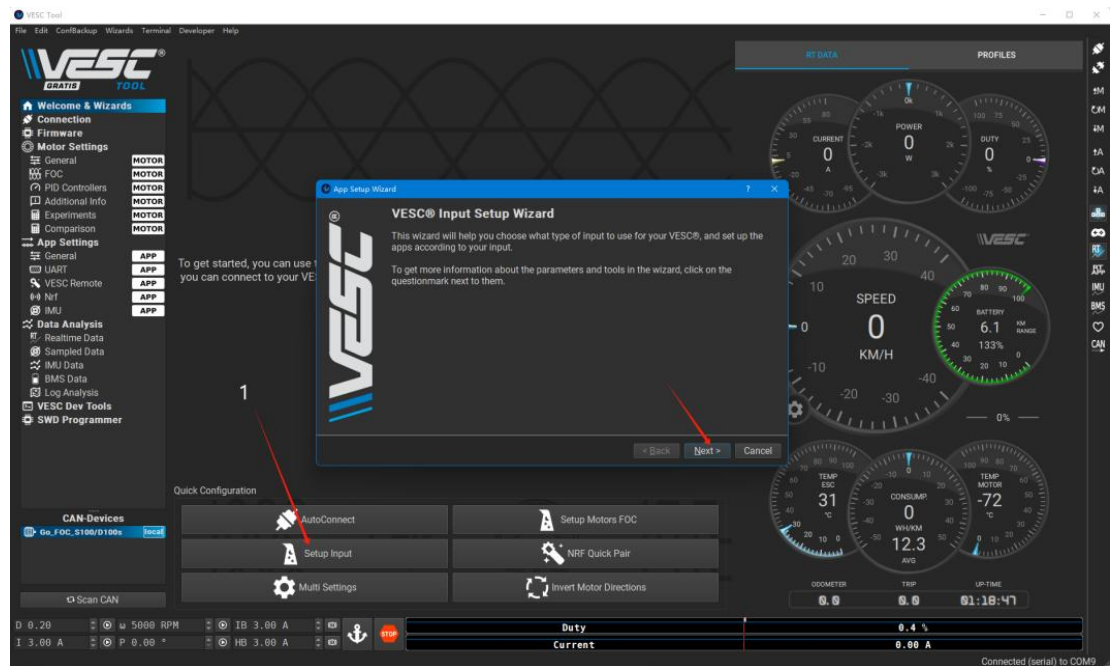


7.Switching the motor direction



4. SIGNAL INPUT

1
Click Setup Input to enter the signal input configuration page.



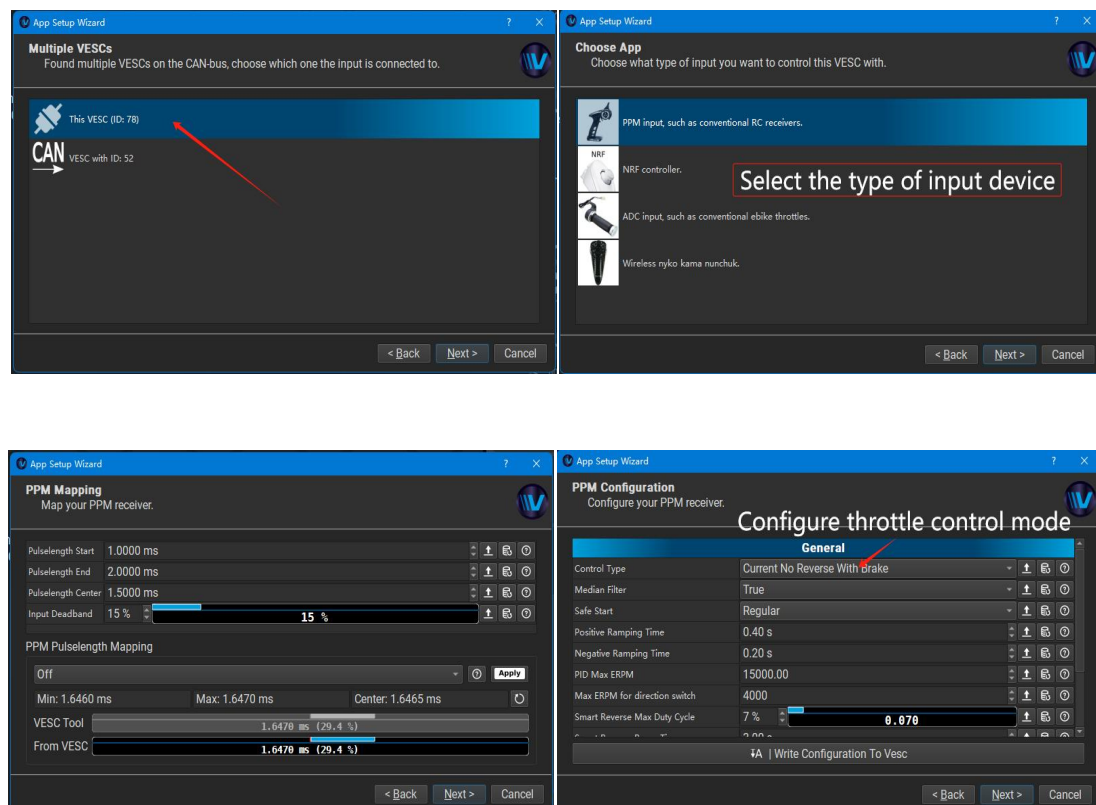
2
Select the port where the receiver or input device is connected.
For example, in the picture, the receiver is installed on the USB side.

3
Select the input device type, such as a PPM receiver.

4
Configure the input signal parameters such as travel range, deadband, etc.

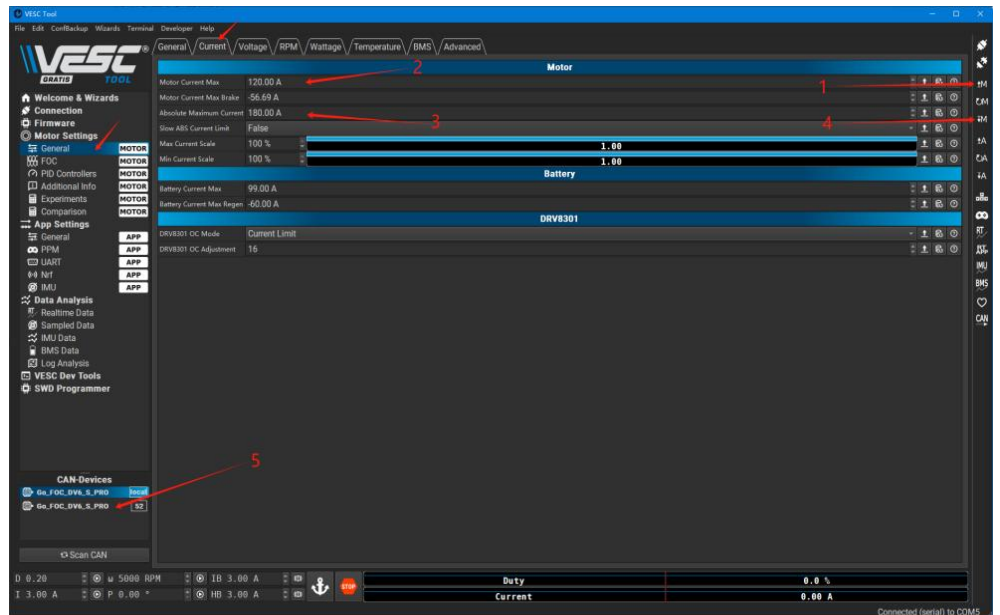
5
Select the control mode, then complete the app input configuration.

At this point, your motor is ready to run!



MODIFY MOTOR CURRENT:

- 1 Click M↑ to read the current motor configuration.
- 2 Modify the motor current.
- 3 Adjust the absolute maximum current (approximately 1.5 times the motor current).
- 4 Click M↓ to write the changes.
- 5 Switch to another ESC and repeat the above steps (M↑ → modify → M↓).

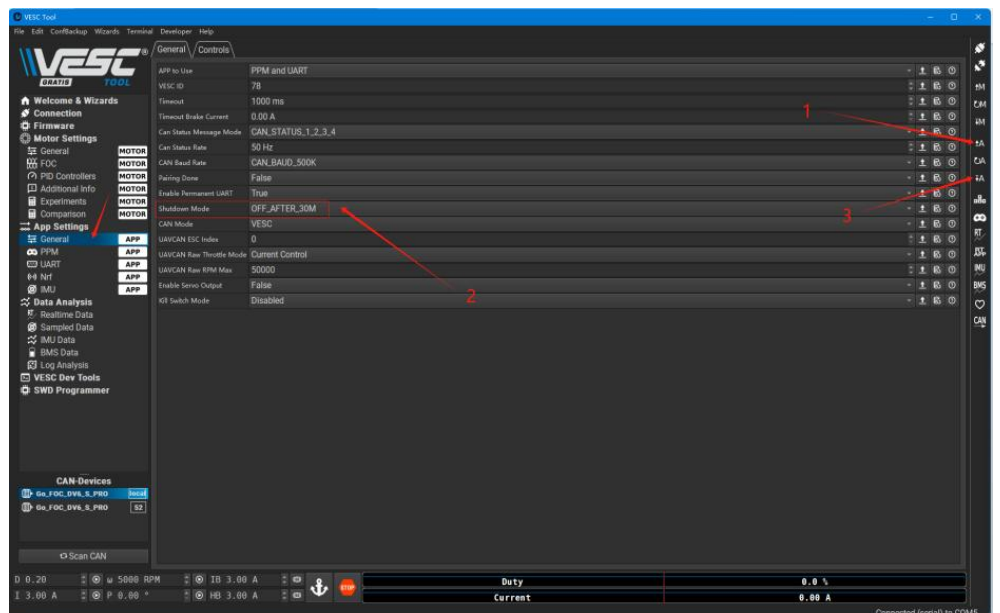


CONFIGURING THE BUILT-IN SLEEP BUTTON:

- 1 Click A↑ to read the APP parameters.
- 2 Modify the shutdown mode, for example, OFF_AFTER_30M, which means the device will automatically power off after 30 minutes of inactivity.
- 3 Click A↓ to write the parameters.

Note: Please verify whether your ESC supports the switch button function before configuring. ESCs with the silkscreen marking "SW" support the switch button feature, usually located near the 8-pin and 3-pin connectors.

For dual-drive ESCs, only the ESC on the SW side needs to have the shutdown mode set; the other ESC should remain ALWAYS ON.



VESC TOOL USAGE TIPS:

Before modifying any parameters, always click Read first, then make your changes, and finally click Write.

M↑ → Modify → M↓

A↑ → Modify → A↓